QUESTION 1

What does "asynchronous" in UART communication imply?

Answer is Correct **Score: 1 / 1**

Options

A. Both systems share the same clock

B. Data is sent without a clock signal

Correct

C. Clock signal is sent with each byte

D. Data and clock are encoded together

FEEDBACK

Solution: UART allows data transfer without a shared clock, hence the term “asynchronous.”

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=101

QUESTION 2

Why can’t a single data wire alone work in UART communication?

Not Attempted **Score: 0 / 1**

Options

A. It doesn’t support voltage signals

Expected

B. There’s no place to ground the data

C. It cannot detect start and stop bits

D. It cannot transmit analog values

FEEDBACK

Solution: A complete circuit is required, and the ground wire provides a common reference

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=215

QUESTION 3

In UART, what is the purpose of the start bit?

Not Attempted **Score: 0 / 1**

Options

A. To increase the baud rate

B. To indicate the end of transmission

Expected

C. To indicate the beginning of data

D. To signal a parity check

FEEDBACK

Solution: The line is pulled low for one bit duration to signal the beginning of data transmission.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=320

QUESTION 4

Which parameters must both UART transmitter and receiver pre-agree upon?

Not Attempted **Score: 0 / 1**

Options

A. Voltage levels and baud rate

Expected

B. Bit duration and baud rate

C. Number of start bits only

D. Number of stop bits only

FEEDBACK

Solution: Both devices must know the bit duration (timing) to correctly interpret each bit.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=374

QUESTION 5

Which of the following can be true about UART data bits?

Not Attempted **Score: 0 / 1**

Options

A. They are always fixed to 8 bits

B. Must be in multiples of 4

Expected

C. Can vary between 5 and 8 bits

D. Only 7-bit data is standard

FEEDBACK

Solution: Though 8 bits is common, UART supports data widths from 5 to 8 bits.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=491

QUESTION 6

What is the purpose of the parity bit in UART?

Not Attempted **Score: 0 / 1**

Options

A. To increase speed

B. To support analog signal transmission

Expected

C. To detect single-bit errors

D. To indicate stop condition

FEEDBACK

Solution: A parity bit is used for simple error detection by checking if the total number of ones is odd or even.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=542

QUESTION 7

What does the UART configuration “8N1” indicate?

Not Attempted **Score: 0 / 1**

Options

A. 8 data bits, no parity, 1 start bit

B. 8 data bits, 1 parity bit, no stop bit

Expected

C. 8 data bits, no parity, 1 stop bit

D. 8-bit address mode with parity

FEEDBACK

Solution: "8N1" refers to 8 data bits, No parity, and 1 stop bit. One start bit is implied by the protocol.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=674

QUESTION 8

What does a baud rate of 9600 mean in UART?

Not Attempted **Score: 0 / 1**

Options

A. 9600 bytes per second

Expected

B. 9600 symbols per second

C. 9600 cycles per second

D. 9600 packets per second

FEEDBACK

Solution: Baud rate refers to the number of symbol changes per second, which for UART equals bits per second.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=721

QUESTION 9

What is the main difference between UART and USART?

Not Attempted **Score: 0 / 1**

Options

A. USART supports USB

Expected

B. USART supports synchronous mode

C. UART supports higher speeds

D. UART uses a shared clock

FEEDBACK

Solution: USART can support synchronous communication. It is based on a shared clock , while UART is asynchronous only means it does not shares clock .

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=862

QUESTION 10

Which mode of UART allows simultaneous two-way communication?

Not Attempted **Score: 0 / 1**

Options

A. Simplex

B. Half duplex

Expected

C. Full duplex

D. Multiplex

FEEDBACK

Solution: Full duplex uses separate TX and RX lines to support simultaneous data transmission and reception.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=923

QUESTION 11

Why is UART not ideal for high-speed or multi-device communication?

Not Attempted **Score: 0 / 1**

Options

A. It requires expensive hardware

B. It uses high voltage

Expected

C. It has no shared clock and high framing overhead

D. It supports only 16-bit data

FEEDBACK

Solution: Lack of a shared clock and per-frame overhead limits its speed and scalability.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=1036

QUESTION 12

Why is UART commonly used for console interfaces?

Not Attempted **Score: 0 / 1**

Options

A. High throughput

Expected

B. Graphical support

C. Simplicity and real-time interactivity

D. Full-duplex error correction

FEEDBACK

Solution: UART is simple, low-cost, and enables real-time debugging through text-based serial communication.

Link: https://youtu.be/RbrNDDwNhPk?feature=shared&t=1273